The Second Derivative, Concavity, & Inflection Points

1. Review: What the First Derivative Tells Us

The first derivative $f'(x)$ gives the instantaneous rate of change of f (slope of the tangent line):

- If $\mathbf{f'(x) > 0}$ on an interval, then $f(x)$ is **increasing** on that interval.
- If $\mathbf{f'(x) < 0}$ on an interval, then $f(x)$ is **decreasing** on that interval.

2. The Second Derivative: Rate of Change of Slope

The **second derivative** is simply the derivative of the first derivative:

$$\mathbf{f''(x) = \frac{d}{dx}\left[f'(x)\right] = \frac{d^2f}{dx^2} = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h}}$$

Just as $f'(x)$ tells us whether $f(x)$ is increasing or decreasing, $f''(x)$ tells us whether the **slopes** $f'(x)$ **themselves** are increasing or decreasing.

Concavity: Geometric & Analytic Characterizations

- **Concave Up ($\cup$):**
  - **Slope Behavior:** Slopes are **increasing** as x moves left to right ($f''(x) > 0$).
  - **Tangent Line Property:** The tangent lines lie **strictly below** the curve.
  - **Intuition:** The graph holds water like a cup.
- **Concave Down ($\cap$):**
  - **Slope Behavior:** Slopes are **decreasing** as x moves left to right ($f''(x) < 0$).
  - **Tangent Line Property:** The tangent lines lie **strictly above** the curve.
  - **Intuition:** The graph sheds water like an umbrella.

3. Crucial Subtlety & Counterexample: $f(x) = x^4$

A common misconception is that concavity requires $f''(x) > 0$ strictly everywhere or that $f''(c) = 0$ implies a change in concavity. Consider:

$$f(x) = x^4 \implies f'(x) = 4x^3 \implies \mathbf{f''(x) = 12x^2}$$

- At $x = 0$, we have $f''(0) = 12(0)^2 = 0$.
- For all $x \neq 0$, $f''(x) = 12x^2 > 0$.
- The slopes $f'(x) = 4x^3$ are strictly increasing across all $x \in \mathbb{R}$ (from negative, through 0, to positive).
- The horizontal tangent line at $(0, 0)$ ($y = 0$) lies strictly below the curve $y = x^4$ for all $x \neq 0$.
- **Conclusion:** $f(x) = x^4$ is **strictly concave up everywhere on** $(-\infty, \infty)$, even though $f''(0) = 0$.

4. Inflection Points

Definition of an Inflection Point:
A point $(c, f(c))$ on the curve is an **inflection point** if:
1. The function f is continuous at c and has a tangent line at $x = c$.
2. The concavity of f **changes sign** across $x = c$ (from concave up to concave down, or vice versa).

Geometric Signature: At an inflection point, the **tangent line cuts across (crosses through)** the curve!

Test for Inflection Points:

1. Find candidate points where $f''(x) = 0$ or $f''(x)$ does not exist.
2. Test test values on intervals to see if $f''(x)$ actually **changes sign** (e.g. from $+$ to $-$ or $-$ to $+$).
3. If $f''(x)$ does not change sign (as in $f(x) = x^4$ where f'' is positive on both sides of 0), it is **not** an inflection point!